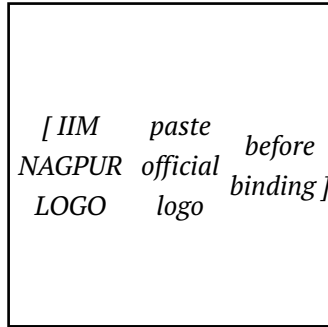


SUMMER INTERNSHIP PROGRAM

Final Project Report



**Commercialisation
Readiness and Valuation
Framework for EIL's
EngSUI Deep-Tech
Portfolio**

Submitted by

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CERTIFICATE

This is to certify that **Mr. Priyanshu Dutta**, a student of the MBA (General Management) 2025-27 batch at the Indian Institute of Management Nagpur, has successfully completed his Summer Internship at **Engineers India Limited (EIL), Gurugram**, from 6th April 2026 to 6th June 2026.

The final report titled "*Commercialisation Readiness and Valuation Framework for EIL's EngSUI Deep-Tech Portfolio*" is a bonafide record of the work carried out by him during this period. The project was executed under the primary guidance of Mr. Sanjeev Singhal, SGM (R&D), and the direct supervision of Mr. Vijay Thakare, Manager (R&D).

During his tenure, Mr. Dutta displayed a high degree of professionalism, analytical rigour, and technical competence. His contribution toward developing a structured deep-tech startup evaluation framework, an accompanying digital platform, and the calibration of both against eight live startup transactions is a substantive addition to the EngSUI initiative's decision-support systems.

We wish him continued success in his academic and professional pursuits.

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My deepest thanks to **Mr. Sanjeev Singhal, SGM (R&D)**, for the intellectual generosity and strategic clarity with which he framed the project brief. His expectation — that the deliverable should be defensible to EIL's investment committee, not merely academically interesting — was the single most important constraint shaping the framework's design.

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The founders and senior personnel of **Fermentech Labs, Powerzest Energy Solutions, Aquafront Infrastructure, E-Spin Nanotech, VayuJal Technologies**, and **Alt Carbon** were generous with their time and candid about their commercialisation barriers. Without their participation, the calibration layer of this report would not exist.

I sincerely thank **Prof. Rakesh Gupta** of IIM Nagpur for academic guidance throughout the internship — particularly for steering Chapter 5's methodology section toward rigour rather than novelty.

Finally, I thank my cohort at IIM Nagpur for sharpening this work through informal critique, and my family for the unwavering encouragement that made eight weeks of focused work possible.

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ABSTRACT

Engineers India Limited (EIL), through its EngSUI initiative, deploys grant and equity capital across deep-tech energy startups but has historically relied on external Transaction Consultant (TC) reports for valuation and on case-by-case judgment for investment rationale. This dependency produces inconsistent comparisons across the portfolio, opaque audit trails, and slow turnaround for investment committee decisions. The problem this internship addressed is the absence of an internal, repeatable framework that quantifies commercial readiness and ties it to a defensible valuation range and a specific engagement pathway.

The deliverable is a three-layer framework — the **Commercialisation Readiness Scorecard (CRS)**, the **Dynamic Weight Calibration System (DWCS)**, and a stage-aware **Scenario Valuation Model** — supported by an interactive browser-based platform deployed at *engsui-evaluator.netlify.app*. The CRS distributes 120 points across seven evaluation modules: Technology Readiness (25), IP & Patents (15), Team & Leadership (20), Market Validation (20), Adoption Feasibility (15), Business Model (15), and EIL Strategic Fit (10). DWCS adjusts module weights by TRL stage, sector maturity, policy environment, IP strength, team credibility, and competitive intensity. The valuation layer auto-selects among Berkus, Scorecard, VC Method, Revenue Multiple, Peer Multiple, and risk-adjusted DCF, and renders Bull / Base / Bear scenarios.

The framework was calibrated against eight live transactions — Alt Carbon, Miko Technologies, E-Spin Nanotech, Log9 Materials, Aquafront Infrastructure, Fermentech Labs, VayuJal, and Powerzest Energy — with five of eight cases landing inside a $\pm 20\%$ variance band against disclosed market valuations. A deliberately included negative case (Miko Technologies) was correctly rejected by the EIL Strategic Fit hard floor despite scoring 79% on the general CRS, validating the EngSUI-specific filter. Calibration revealed three sector multiples requiring adjustment (CCUS, nano, wind) and prompted introduction of an "embryonic" DCF tier for pre-revenue moonshots — both now reflected in the production tool.

Five recommendations are offered to EIL: (i) formally adopt the CRS platform as the standard pre-screen for EngSUI applications; (ii) make DPIIT recognition a prerequisite for Direct Equity tracks; (iii) embed pro-rata

participation and information rights in all future MOAs; (iv) institute mandatory IP-counsel review above ₹5 Cr commitment; and (v) build a lightweight API bridge to MCA21 and the Startup India portal for automated compliance verification. The framework, platform, and calibration corpus together constitute a transferable internal capability that reduces TC dependency, improves cross-portfolio comparability, and creates an auditable decision trail aligned with the constraints of a Navratna PSU investor.

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CHAPTER 1 – AIM AND ESTABLISHMENT OF THE COMPANY

1.1 About EIL

Engineers India Limited (EIL) is a Navratna Public Sector Undertaking under the Ministry of Petroleum and Natural Gas (MoPNG), Government of India, established in 1965. Headquartered at *Engineers India Bhawan*, R. K. Puram, New Delhi, with its primary technical and project execution operations at the Gurugram complex on NH-8, EIL has, over six decades, become the country's principal engineering consultancy and Engineering-Procurement-Construction Management (EPCM) services provider for the hydrocarbon and infrastructure sectors. The company employs over 2,800 professionals, of whom approximately eighty percent are technically qualified engineers and scientists.

EIL's revenue base of roughly ₹3,500 Cr (FY24) is derived predominantly from consultancy and turnkey project execution mandates. The company has executed projects across more than thirty countries spanning the Middle East, Africa, and South-East Asia, in addition to a dominant position in domestic refinery, gas processing, and pipeline projects. Of particular relevance to this internship, EIL has been progressively widening its mandate from conventional hydrocarbons into the green-energy and sustainability domains — a transition that creates the strategic logic for the EngSUI initiative discussed in Chapter 2.

1.2 Position in Industry

EIL operates as a technical-services intermediary between the Government of India's energy-policy mandates and the operational execution of large capital projects across the public-sector hydrocarbon ecosystem. Its primary clients include ONGC, Indian Oil Corporation Limited (IOCL), Bharat Petroleum Corporation Limited (BPCL), Hindustan Petroleum Corporation Limited (HPCL), GAIL, Numaligarh Refinery, and Mangalore Refinery and Petrochemicals (MRPL). The company has also developed specialised engineering capabilities for the fertiliser sector, metallurgy, and increasingly for renewable energy deployment.

In the green-energy transition, EIL is positioned as the technical integrator of choice for the National Green Hydrogen Mission, Compressed Bio-Gas (CBG) infrastructure under the SATAT scheme, ethanol blending capacity expansion, biofuel refinery retrofits, and several Carbon Capture Utilisation and Storage (CCUS)

demonstration projects. This strategic repositioning — from a hydrocarbon-EPC firm to a pan-energy technology integrator — directly motivates EIL's interest in early engagement with deep-tech startups, which is the focus of the EngSUI programme and of this project.

EIL's competitive position rests on four reinforcing assets: a six-decade process-engineering knowledge base; access to public-sector procurement cycles unavailable to private engineering firms; a captive talent pipeline through its sponsored institutions; and a balance-sheet capacity that supports turnkey commitments at scale. For deep-tech startups, an engagement with EIL therefore offers a value proposition that is materially different from a financial-investor relationship: it offers route-to-market, not just capital.

1.3 Services and Business Segments

EIL's revenue is broadly attributable to two operating segments: Consultancy & Engineering Projects and Turnkey Projects. Within these, six service lines structure the company's day-to-day delivery: Project Management Consultancy (PMC), Engineering-Procurement-Construction Management (EPCM), Turnkey Engineering Procurement and Construction (EPC), Technology Licensing, Research and Development, and specialised Advisory services for public-sector energy transition.

Of these, the R&D function — within which this internship was housed — is among the smaller in headcount but disproportionately strategic in influence. The R&D Division at Gurugram is the custodian of EIL's indigenous process know-how, proprietary catalyst formulations, and the EngSUI initiative that connects EIL with the external deep-tech ecosystem. The division operates within the constraints of a PSU governance structure — board approvals, public-procurement norms, CAG audit exposure — and these constraints in turn shape what a "valuation framework" must look like for EIL to use it credibly.

1.4 Mission and Strategic Direction

EIL's institutional mission, captured in its tagline "Delivering Excellence Through People," has in recent years been reframed around three axes: Net-Zero alignment, indigenisation of process technology, and acceleration of deep-tech commercialisation in the Indian energy sector. The

first axis is policy-driven, the second is competitive, and the third — addressed directly by EngSUI — is structural. The argument is that India will not achieve its 2070 net-zero commitment if commercialisation timelines for indigenous deep-tech energy innovations continue to stretch beyond a decade. EngSUI exists to compress that timeline by leveraging EIL's EPC network as a deployment partner for startup-stage technologies.

This strategic framing is consequential for the project at hand. A valuation framework designed for a generalist venture capital fund would optimise for financial-return maximisation. The framework developed in this internship instead has to optimise for *EIL-specific strategic deployment value* — which includes EPCM relevance, sector alignment, EIL's role beyond capital, and the operational risk that a Navratna PSU can responsibly underwrite. This distinction is the single most important design constraint shaping Chapter 5.

CHAPTER 2 — POLICY OF THE COMPANY

2.1 Innovation and Technology Transfer Policy

EIL's innovation policy operates within the framework of the Department of Public Enterprises (DPE) guidelines for R&D investment by Navratna enterprises, the DSIR recognition of EIL's R&D Centre, and the Atmanirbhar Bharat indigenisation mandate. Under these instruments, EIL invests approximately one to two percent of its annual turnover into R&D activities, with a stated objective of building defensible intellectual property and reducing import dependence on foreign process licensors.

The technology-transfer policy distinguishes between three modes of external engagement. The first is collaborative research with academic institutions — historically with IITs, IISc, and CSIR labs. The second is licensing inward, where EIL acquires use-rights to externally developed process technology to bundle into EPC offerings. The third, and most recent, is direct equity engagement with deep-tech startups through EngSUI — the focus of this report.

For the purposes of the framework developed here, two governance constraints from this policy environment are material. First, every EngSUI investment decision is auditable by the Comptroller and Auditor General of India (CAG), which means rationale and supporting evidence must be captured and traceable. Second, valuation conclusions arrived at internally must be reconcilable with TC reports prepared by registered valuers, since the latter are the legal anchor for any equity transaction. The framework therefore aims to complement, not replace, the TC report — by providing the strategic rationale that TC numbers cannot.

2.2 EngSUI — EIL's Startup India Initiative

EIL's Startup India Initiative — abbreviated as EngSUI — was launched in 2017 in response to the Government's Startup India mission and the MoPNG's 2017 directive that public-sector energy enterprises actively support indigenous deep-tech innovation. EngSUI is administered within the R&D Division at Gurugram and operates as a three-track funding instrument linked to a network of academic host institutions.

EngSUI's stated objectives are to (i) accelerate the commercialisation of deep-tech energy innovations in India; (ii) provide non-dilutive grants and stage-appropriate equity capital where conventional financing is unavailable; (iii) bridge the lab-to-commercial gap by leveraging EIL's EPCM relationships as a route-to-market; and (iv) build a portfolio of strategically aligned technologies that EIL can deploy in client projects. Operational accountability for the programme sits with the SGM (R&D), with day-to-day management by a Manager-level officer and a working-level Engineer/Officer who liaises directly with founders.

2.3 EngSUI Funding Tracks and Host Institution Network

EngSUI deploys capital through three structured tracks differentiated by startup maturity, ticket size, and eligibility. Table 2.1 summarises the three tracks; Table 2.2 lists the host institution network as of June 2026.

Table 2.1 EngSUI funding tracks and eligibility

Track	Maximum Ticket	Eligibility	Stage Target
Innovation Challenge	₹20 Lakh	Any Indian-resident innovator (student or non-student); novel idea-to-prototype work.	Concept to early prototype (TRL 2–4)
Seed Fund	₹100 Lakh	DPIIT-certified startup with demonstrated proof-of-concept; founder-incubator co-application via host institution.	Prototype to pilot (TRL 4–6)
Direct Equity	₹500 Lakh	DPIIT-certified startup with marketable product/process; verifiable revenue traction; cap-table cleanliness.	Pilot to commercial (TRL 7–9)

Eligibility for the Innovation Challenge is broad and treats the grant as risk capital for early experimentation. The Seed Fund presumes a working prototype and ties the deployment to a sponsoring host institution. The Direct Equity route — the most consequential for valuation purposes — is reserved for startups with audited financials, DPIIT recognition, and sufficient commercial traction to justify a priced equity round. The

framework developed in this internship is most directly applicable to the Direct Equity and Seed Fund tracks; for the Innovation Challenge, only the Technology and EIL Strategic Fit modules are typically meaningful.

Table 2.2 EngSUI host institution network (selected, 2017–2026)

Institution Class	Hosts	Representative Programmes
IITs	Ropar, Mandi, Roorkee (TIDES), Madras (IIT-M Incubation Cell), Guwahati, Kanpur (SIIC), Bombay (SINE), Gandhinagar, (BHU MCIIE)	Process tech, energy materials, AWG, AI for industry
NITs	Meghalaya, Jalandhar, Bhopal, Nagpur (CIVN)	Bio-energy, biofuels, instrumentation
Other	a-IDEA NAARM (BioNest), CSIR labs	Industrial biotech, agri-bio

Since 2017, EngSUI has run seven funding rounds. As of mid-2026, the portfolio comprises thirty-one startups funded, of which thirteen have completed milestones, fourteen are in progress, and four are pending Memorandum of Agreement execution. This is the population to which the framework presented in Chapter 5 is intended to apply.

2.4 Focus Areas and Portfolio Composition

EngSUI's focus areas are explicitly aligned with EIL's downstream operational footprint and the Government's energy-transition priorities. The six declared focus areas are: hydrocarbon process technology; oil and gas hardware manufacturing; IT and Industrial IoT for energy operations; renewable energy and storage; biofuels, waste-to-energy, and nanomaterials; and Net-Zero and HSE technologies. The portfolio composition — observed during the desk-review phase of this internship — skews toward water tech, biotechnology, and renewables, with underrepresentation in CCUS and nuclear-adjacent SMR technologies.

For the purposes of the framework, three implications follow. First, the EIL Strategic Fit module must encode *which* focus area a startup addresses, not just whether it is broadly "deep-tech." Second, the sector multiples used in the valuation engine must reflect the actual exit multiples observed in

each focus area, not a single tech-grade default. Third, calibration must include at least one negative case from outside the focus areas to test the rejection logic — a role played in this project by Miko Technologies, discussed in Chapter 5.

CHAPTER 3 – ORGANISATIONAL STRUCTURE OF THE COMPANY

3.1 Functional Divisions and Reporting Hierarchy

EIL is organised across five primary functional verticals: Engineering Divisions (Process, Piping, Civil, Electrical, Instrumentation, Mechanical, Environmental); Project Execution; Business Development; Research and Development; and Corporate Services (Finance, HR, Legal, Company Secretariat). Each vertical is led by an Executive Director or General Manager and reports through a structured Director-level interface to the Chairman and Managing Director.

The internship was hosted within the Research and Development Division, which itself reports to the Director (Technical). Within R&D, the EngSUI programme is owned by the Senior General Manager (R&D) and operationalised by a Manager-level officer responsible for portfolio monitoring, due diligence coordination, and external interface with host institutions and founders. This compact reporting line — three levels from the Director — is what makes EngSUI capable of moving at startup- relevant timescales despite operating inside a PSU governance envelope.

3.2 R&D Division and Project Context

The R&D Division at Gurugram is custodian of EIL's proprietary process know-how across hydrogen, refining, petrochemicals, and increasingly the emerging energy-transition technologies. It is also the point of corporate interface with the academic ecosystem: collaborative research agreements, sponsored chairs, and the EngSUI initiative are all administered here. The division's mandate has shifted materially over the past decade from technology-acquisition (licensing inward) toward technology-creation and technology-commercialisation, of which EngSUI is the most public manifestation.

Within this division, the project was scoped as a six-credit MBA Summer Internship deliverable with a working brief shared on Day 1: develop a structured framework that EIL R&D can use to evaluate the commercialisation potential of EngSUI startups, complete with a tool that

operationalises the framework and a calibration exercise against live transactions. The framework was to be defensible to a future investment-committee audience and reusable across multiple funding rounds.

3.3 Key Personnel for the Project

Three individuals at EIL and one at IIM Nagpur formed the immediate project context. Their roles are summarised in Table 3.1.

Table 3.1 Key stakeholders for the project

Name	Designation	Role in Project
Mr. Sanjeev Singhal	SGM (R&D), EIL Gurugram	Primary Mentor — scope definition, weekly framework reviews, framing of the investment-committee defensibility test.
Mr. Vijay Thakare	Manager (R&D), EIL Gurugram	Project Supervisor — day-to-day technical guidance, calibration data access, founder-interview coordination.
Mr. Saurabh Agarwal	Working contact, EngSUI portfolio	Operational interface for startup access, document review, internal documentation alignment.
Prof. Rakesh Gupta	Faculty Guide, IIM Nagpur	Academic guide — methodology rigour, literature anchoring, final-report review.

Six EngSUI founders or senior personnel — across Fermentech Labs, Powerzest Energy Solutions, Aquafront Infrastructure, E-Spin Nanotech, VayuJal Technologies, and Alt Carbon — participated in the structured interview programme described in Chapter 5.10. Their contribution constitutes the primary-research substrate for the calibration layer.

CHAPTER 4 – WORK EXPERIENCE

4.1 Onboarding and Orientation

The internship began on 6 April 2026 with a one-week onboarding sequence at the EIL Gurugram complex. The first three days were spent on administrative formalities, security clearance, and an overview of EIL's corporate structure delivered through HR-led sessions. From Day Four, the orientation shifted to project-specific induction: a closed review of three sample TC reports (with sensitive financial figures redacted), a walkthrough of the EngSUI portfolio dashboard maintained by the R&D Division, and an introductory meeting with the SGM (R&D) where the scope and acceptance criteria for the framework were set.

Two confidentiality undertakings shaped the rest of the engagement. First, no TC report content could be cited verbatim outside the project; the framework therefore had to be designed from publicly inferable patterns rather than from confidential TC inputs. Second, proprietary EngSUI startup financials could be referenced only in indexed, ratio-converted, or qualitative form. The calibration layer of the framework was built within these constraints, using publicly disclosed valuations (Tracxn, MCA21, press coverage) as anchors.

4.2 Key Concepts Encountered

The internship required rapid absorption of several specialised concepts that intersect strategy, finance, and engineering. The most consequential are summarised below; each is treated in greater depth in the relevant sub-section of Chapter 5.

Technology Readiness Level (TRL). The nine-step European Commission TRL scale measures the maturity of a technology from basic principles observed (TRL 1) through successful commercial deployment (TRL 9). For EIL, TRL serves both as a screening tool and as a driver for valuation methodology selection — a TRL 3 startup should not be valued with the same instrument as a TRL 8 operator, a point formalised in the DWCS and the Stage-to-Method bridge.

EPCM Mandate. Engineering-Procurement-Construction Management is the operating model through which EIL delivers value to end clients. A deep-tech startup's relevance to EIL is materially enhanced if its technology can be integrated into an EPCM scope of work, since that creates a deployment route that is closed to financial investors. This is encoded in the EIL Strategic Fit module under the EPC sub-indicator.

PSU Procurement Norms. Public-procurement constraints — vendor qualification, BIS/IS certifications, three-bid systems, indigenous content requirements — operate as adoption hurdles for startups that propose to sell into the PSU client base. The Adoption Feasibility module incorporates these constraints rather than treating them as obstacles outside the startup's control.

Deep-Tech Commercialisation Gap. NESTA's 2022 analysis of hardware-startup failures showed that approximately forty percent fail not at the proof-of-concept stage but at the scale-up stage, where capital intensity, manufacturing maturity, and adoption economics collide. This empirical finding directly justifies the twenty-five-point weight allocated to the Technology Readiness module.

Pre-Revenue Valuation Methods. The Berkus, Scorecard, and Risk-Factor Summation methods were studied to understand how angel investors price companies without revenue. The Berkus method's five-milestone structure, in particular, informed the design of the confidence-flag layer in the framework.

DPIIT Recognition and Angel Tax. Eligibility for EngSUI Seed and Direct Equity tracks is conditional on DPIIT recognition. DPIIT-recognised startups are also exempt from Angel Tax on premium-priced equity rounds, a non-trivial structuring consideration that the framework flags via the Data Room module.

Staged Equity Conversion. EngSUI's instruments sometimes structure capital as Compulsorily Convertible Debentures (CCDs) or staged equity tranches. Understanding the underlying instrument is necessary to translate a "valuation" number into a correctly priced engagement.

4.3 Organisational Learning

Beyond technical content, the internship surfaced a set of organisational realities specific to PSU-investor behaviour that materially shape framework design. Four observations stand out.

First, EIL's investment decisions are governed less by IRR optimisation than by strategic deployment value combined with audit defensibility. A framework that produces an attractive return number but cannot be traced back to underlying evidence is unusable; the inverse — modest returns with full evidence chain — is preferred.

Second, EIL's evaluation today is heavily TC-dependent. TC reports deliver valuation numbers but not investment rationale — they answer "what is it worth?" but not "why should EIL engage?" The framework therefore positions itself as a rationale layer above the TC number, not a replacement for it.

Third, BIRAC's grant-worthiness framework was studied as a comparative reference. BIRAC evaluates whether a biotech startup deserves non-dilutive grant capital; EIL needs to evaluate whether a startup is a commercial partner — a fundamentally different question. The CRS is deliberately structured around the latter.

Fourth, the transition from a grant-led EngSUI model toward direct equity engagement is institutionally significant. It implies different governance, different exit considerations, and a tighter standard for cap-table cleanliness and IP clarity. The Data Room module of the platform was designed in response to this transition.

4.4 EngSUI Portfolio Exposure

Across eight weeks, structured exposure was obtained to the active EngSUI portfolio through three mechanisms: a desk review of the portfolio brochure and MOAs, a guided walkthrough of the internal portfolio dashboard, and direct interviews with six founders coordinated through the R&D Division. The six founder interviews are the primary research backbone of this report and are documented in detail in Chapter 5.10.

The portfolio's diversity was educational in itself. It includes industrial biotech (Fermentech's solid-state fermentation enzymes), micro-renewable energy (Powerzest's Adaptable Disc Turbine), water infrastructure (Aquafront's floating CNG and modular pontoons), nanomaterial-based filtration (E-Spin's electrospinning equipment), atmospheric water generation (VayuJal's IIT-Madras spinout), and Enhanced Rock Weathering for carbon removal (Alt Carbon). Each presents a different combination of TRL, capital intensity, market structure, and EIL fit — making the portfolio an ideal calibration set for a framework that must work across deep-tech heterogeneity.

4.5 Macro-environmental Factors Shaping the Project

Three macro factors materially shaped both the project framing and the framework's design choices. First, India's commitment to net-zero by 2070 has converted what was previously a niche climate-tech investor consideration into a mainstream public-policy mandate, increasing the political pull for deep-tech energy commercialisation. Second, global decoupling from Chinese clean-energy supply chains creates a direct industrial-policy case for indigenous manufacturers in solar, energy storage, and electrolyser technologies — sharpening the policy sub-indicator within Market Validation. Third, the National Green Hydrogen Mission and the Compressed Bio-Gas (SATAT) framework provide explicit demand-side anchors for several EngSUI startups, which the framework allows analysts to capture through both the Market and EIL Fit modules.

These macro tailwinds do not, however, substitute for startup-specific diligence. A common analytical error during the calibration phase was over-weighting policy tailwind into individual-startup CRS scores. The DWCS partially corrects for this by adjusting the policy weight at sector level rather than at startup level — the rationale is detailed in Chapter 5.6.

CHAPTER 5 — DESCRIPTION OF WORK RESPONSIBILITIES TAKEN

5.1 Problem Statement

EIL's EngSUI initiative has, in nearly a decade of operation, accumulated thirty-one funded startups, multiple host-institution relationships, and a sectoral footprint that spans water, biotech, storage, hydrogen, CCUS, and waste-to-energy. Yet the institutional decision process by which an EngSUI applicant moves from initial screening to a funded engagement remains substantially case-by-case. There is no internally owned, repeatable framework that quantifies a startup's commercial readiness, ties that quantification to a defensible valuation range, and recommends a specific engagement pathway (equity, licensing, engineering partnership, or watch). Five consequences follow from this gap.

First, comparability across the portfolio is weak. When two startups are evaluated by different TCs at different points in time, the resulting valuation numbers cannot be meaningfully ranked. Second, rationale capture is shallow. TC reports deliver valuation numbers but not the strategic basis on which EIL is choosing to engage — information that is essential for CAG-defensibility and for investment-committee deliberation. Third, turnaround is slow. The dependency on external TCs introduces calendar delays inconsistent with startup operating timescales. Fourth, BIRAC-style frameworks, where available, evaluate grant-worthiness rather than commercial partnership worthiness; they are conceptually misaligned with the Direct Equity track. Fifth, the absence of a calibrated, EIL-specific method discourages internal capability-building and entrenches TC dependency.

The problem this project addresses is precisely this gap: the absence of a framework that is internal, repeatable, calibrated, EIL-specific, auditable, and operationalised as a working tool. Each adjective is deliberate. "Internal" excludes outsourcing to TCs. "Repeatable" rules out one-off bespoke analyses. "Calibrated" requires the framework to have been tested against real transactions. "EIL-specific" enforces the strategic-fit primacy that distinguishes EIL from a generalist VC. "Auditable" satisfies the CAG defensibility requirement. And "operationalised" — i.e. delivered as a platform, not a slide deck — converts an analytic framework into an actually-usable tool.

5.2 Scope and Objectives

The project's scope was defined at the outset by the SGM (R&D) in the Day-One scoping conversation and confirmed in writing in Week 2. The scope covers (i) deep-tech startups in EIL's declared focus areas, (ii) all three EngSUI tracks but with particular emphasis on Seed Fund and Direct Equity, (iii) the design of a scoring framework, a weight calibration system, and a stage-appropriate valuation methodology, (iv) the build of a digital platform that operationalises these components, and (v) calibration against a minimum of three live transactions with publicly disclosed valuations. The scope explicitly excludes (i) consumer-tech, services-only, or pure-software startups, (ii) public-equity investments, (iii) M&A advisory, and (iv) creation of legal documentation for any specific transaction.

Within this scope, six objectives were committed to:

1. Review valuation methodologies used by external TCs across the EngSUI portfolio and identify their structural limitations as decision instruments for a strategic PSU investor.
2. Design a Commercialisation Readiness Scorecard (CRS) — a quantitative framework that converts startup attributes into a single comparable score, accompanied by an EIL Strategic Fit overlay that acts as a hard filter.
3. Construct a Dynamic Weight Calibration System (DWCS) that adjusts module weights by TRL stage, sector maturity, policy environment, IP strength, team credibility, and competitive intensity.
4. Build a Scenario Valuation Model that applies the right valuation method by stage and produces Bull, Base, and Bear valuations with auditable underlying assumptions.
5. Develop and deploy a browser-based platform that operationalises the framework end-to-end — from data capture through scoring, valuation, and recommendation generation.
6. Calibrate the framework against at least three live transactions, expanding to additional cases where time permitted, and document variance with a transparent diagnostic narrative.

The deliverable bundle, accordingly, comprises this final report; the deployed platform at *engsui-evaluator.netlify.app*; the underlying scoring hierarchy reference; the platform functioning guide; the calibration dataset for eight startups; and a portfolio EIL Role matrix derived from the calibration exercise.

5.3 Research Methodology

The methodology combined primary research (structured founder interviews), secondary research (regulatory filings, transaction databases, academic literature), and applied analytical methods (TRL assessment, TAM/SAM/SOM sizing, Porter's Five Forces). The sequencing was deliberately iterative: framework design preceded interviews so that interviews could test specific design hypotheses; calibration followed interviews so that observed market behaviour could anchor parameter choices; framework refinement followed calibration so that observed variance could be diagnosed and addressed.

Primary research.

A structured thirty-two-question interview instrument was designed across five modules — Technology, Market & Customers, Financials & Scale, EIL Partnership, and Risks & Barriers. The instrument is reproduced as Appendix B. Interviews were conducted with six founders or senior personnel between Weeks 5 and 7; each interview ran forty to seventy-five minutes, was documented in a structured form, and was cross-referenced with publicly available data prior to coding into the platform. Interview coverage is summarised in Table 5.7.

Secondary research.

Five sources structured the secondary corpus: (i) MCA21 filings for incorporation, paid-up capital, directorships, and AGM compliance status; (ii) the Tracxn Pro deep-tech database for funding rounds, post-money valuations, and competitor mapping; (iii) Tofler for filed annual financials; (iv) credible press coverage (Entrackr, Inc42, TechCrunch, YourStory) for round-level detail; and (v) BIS, CPCB, and DPIIT registries for certification and recognition data. Academic anchors include Berkus (1996), Payne (2001), Sahlman (1990), Porter (1980), Blank (2013), NESTA (2022), CB Insights (2021), and Europ (2025).

Analytical frameworks.

The applied analytical layer combines Porter's Five Forces for competitive analysis, the European Commission's nine-level TRL scale, TAM/SAM/SOM market sizing, Steve Blank's Customer Development methodology adapted for PSU constraints, and Bull/Base/ Bear scenario construction. The integration logic is described in sub-sections 5.5 through 5.7.

5.4 Limitations of Existing Methodologies

Three families of methodology were reviewed prior to framework design: angel/seed-stage methods (Berkus, Scorecard, Risk Factor Summation), pre-revenue venture methods (VC Method, First Chicago, Sahlman), and revenue-stage methods (DCF, Revenue Multiple, Comparable Transactions). Each is useful within its own stage band but each carries structural limitations when applied to EIL's EngSUI context.

The **Berkus method** assigns up to USD 500,000 per risk-reduction milestone across five domains (idea, prototype, team, strategic relationships, sales rollout). It is useful for TRL 1–4 work but has two problems for EIL. First, the absolute cap was calibrated for the US angel ecosystem; without re-anchoring, the Indian deep-tech equivalent under-shoots. Second, the method lacks any explicit strategic-fit weighting, which is precisely the dimension EIL needs to evaluate.

The **Scorecard method** (Payne, 2001) adjusts a median pre-money valuation in the relevant geography against six factors — Strength of Team, Size of Opportunity, Product, Sales Channels, Stage of Business, and Other. It is more sophisticated than Berkus but inherits two related weaknesses: it presumes a well-defined regional median, and it uses fixed weights regardless of startup stage. Both presumptions fail in Indian deep-tech, where medians are thin and stage variance is extreme.

The **VC Method** (Sahlman) projects an exit value at a target year and discounts back at the investor's required IRR. It is well suited to pre-revenue ventures with credible projection horizons but is sensitive to terminal-multiple assumptions. The framework adopted here uses VC Method as the primary instrument in the 70–84% CRS band but anchors the terminal multiple to a sector-specific table rather than a tech-grade default.

The **DCF method** is the rigorous instrument of choice for revenue-generating businesses but historically over-discounts pre-revenue deep-tech because TRL risk is layered into the WACC without a corresponding adjustment in the terminal multiple. The framework partitions DCF into three tiers — mature, projecting, and embryonic — each with a stage-appropriate WACC and terminal multiple. Calibration revealed that the original 35%/6× defaults overshoot pre-revenue micro-renewables, leading to the introduction of an explicit embryonic tier at 40%/4×.

The **Revenue Multiple** method is anchored on trailing or forward ARR scaled by a sector multiple. It is the cleanest instrument for revenue-generating early growth companies but requires a defensible sector multiple. The framework includes a fourteen-sector multiple table (Table 5.6), calibrated against observed exit multiples and adjusted in line with calibration variance.

A common limitation across all five methods is the absence of a *strategic-fit hard gate*. For a generalist VC, strategic fit is a soft preference. For a Navratna PSU with a specific energy- transition mandate, strategic fit is the difference between an eligible investment and an out-of-mandate one. The framework therefore introduces an EIL Strategic Fit module as a hard floor — a CRS < 50 reject, an EIL Fit < 3 reject, and a Technology < 10 reject — that no aggregate score can override.

5.5 The Commercialisation Readiness Scorecard (CRS)

The CRS distributes 120 points across seven evaluation modules. The seven-module structure was chosen to align with the standard consultant due-diligence rubric (Client, Company, Sector, Business Plan, Financials, Valuation, Legal, Recommendation) while re-weighting toward deep-tech-specific risks. Table 5.1 summarises the module overview; the sub-indicators within each module are enumerated in Appendix A.

Table 5.1 CRS module overview

Module	Pts	%	Key Indicators	Empirical Anchor
M1 — Technology Readiness	25	21%	TRL level, pilot status, independent validation, data quality, uniqueness, scalability, manufacturing readiness, reliability	~40% of hardware startups fail at scale (NESTA 2022)
M2 — IP & Patent Assessment	15	13%	Filing status, claim breadth, claim quality, FTO, publications, ownership/origin, trade-secret depth	Non-patented tech commoditises once funding is announced
M3 — Team and Leadership	20	17%	Founder domain expertise, business depth, network capital, completeness, institutional anchoring, diversity, advisory bench, prior traction, full-time intensity	Team quality predicts outcome better than idea quality (YC, Sequoia, Bessemer)
M4 — Market Validation	20	17%	TAM, SAM, CAGR, customer discovery depth, LOI/pilot evidence, paying customers, unit-of-consumption fit, competitive intensity, policy tailwind	~35% of startups fail from no market need (CB Insights 2021)
M5 — Adoption Feasibility	15	13%	Regulatory approval path, CapEx requirement, integration complexity, incumbent disruption, supply-chain readiness, price competitiveness	Deep-tech has the longest lab-to-scale gap of any sector
M6 — Business Model	15	13%	Revenue model clarity, unit economics, customer acquisition strategy, operating discipline, ARR trajectory, runway	Revenue-model ambiguity is the 2nd most common failure mode (CB Insights)
M7 — EIL Strategic Fit	10	8%	Sector alignment, EPC integration potential, National-Mission fit,	EIL has a sector-specific mandate, not a generalist VC mandate

			licensing willingness, founder receptivity to EIL involvement	
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The module weights are not asserted as universal truths — they are deliberate design choices that reflect EIL's mandate and the empirical failure-mode literature. Technology is weighted highest because the empirical evidence on hardware-startup failure concentrates at the scale-up gate. Team is second because longitudinal studies across YC, Sequoia, and Bessemer portfolios repeatedly show that team quality dominates idea quality in explaining outcome variance. Market Validation matches Team because absence of demand is the single largest failure cause. The remaining three modules — IP, Adoption, Business Model — are weighted at fifteen points each as they each capture a distinct risk dimension. EIL Strategic Fit is weighted lowest in points but operates as a hard floor: it filters rather than scores.

5.5.1 Scoring Bands and Engagement Recommendations

Table 5.2 CRS score bands and engagement recommendation

Score Range	%	Band	Recommended Engagement
≥ 102	≥ 85%	Investment-grade	Direct Equity (subject to gates)
90 – 101	75 – 84%	Commercially ready	Equity / Licensing (gates)
70 – 89	58 – 74%	Promising — validate	Engineering Partner / Seed Fund
50 – 69	42 – 57%	High risk — conditional	Watch / Innovation Challenge
< 50	< 42%	Not commercially viable	Reject

Table 5.3 Engagement decision gates

Engagement	CRS Threshold	Module Gates	Additional Conditions
Equity	≥ 85	EIL Fit ≥ 7; EPC Fit ≥ 3	Founder receptivity to EIL involvement
License	≥ 70	IP ≥ 10; EIL Fit ≥ 5	Willingness to license the IP to EIL or its clients
Engineering Partner	≥ 55	EPC Fit ≥ 3; EIL Fit ≥ 4	Compatible technology integration into EPC scope
Watch	42 – 55%	—	Re-evaluate in next funding round
Reject (Hard Floor)	< 50	Tech < 10 OR EIL Fit < 3	Any single hard-floor breach forces reject

The hard-floor logic is the framework's most consequential design decision. An overall CRS of 79% — as observed in the Miko Technologies calibration case — does not unlock EngSUI engagement if the EIL Strategic Fit module scores below three. This deliberately filters out commercially excellent startups that fall outside the EngSUI mandate. The Miko case in Section 5.11 confirms that the hard floor functions as intended.

5.6 Dynamic Weight Calibration System (DWCS)

The CRS uses a fixed weight allocation by default. The Dynamic Weight Calibration System is the second-lens overlay that re-weights modules contextually so that the framework can read a TRL 3 lab-stage startup differently from a TRL 8 commercial operator. DWCS is therefore not a competing score; it is a contextual reading of the same CRS that surfaces when stage, sector, or policy conditions shift the analytically relevant weights.

Table 5.4 DWCS multiplier matrix by TRL stage

Stage	Tech	IP	Team	Market	Adoption	Business	EIL
Early (TRL 1–3)	1.3	1.1	1.1	0.7	0.7	0.5	1.0
Mid (TRL 4–6)	1.0	1.1	1.1	1.0	1.0	0.9	1.0
Late (TRL 7–9)	0.8	0.9	1.0	1.2	1.2	1.3	1.0

The DWCS formula is:

$$DWCS-CRS = (\sum moduleScore \times multiplier) / (\sum moduleCap \times multiplier) \times 120$$

Five additional calibration variables modulate weights beyond TRL stage. *Sector maturity*: in a nascent sector with no incumbents, Market Validation weight rises (market existence must be proven); in a mature sector with strong incumbents, Competitive Position weight rises (differentiation must be proven). *Policy environment*: a strong tailwind (e.g. National Green Hydrogen Mission) increases Industry Analysis weight, since policy de-risks adoption. *IP strength*: a granted patent portfolio increases Technology weight; the absence of IP shifts weight onto Competitive Position. *Team credibility*: a prior exit or domain PhD increases Team weight. *Competitive intensity*: high intensity (multiple well-funded incumbents) increases Competitive Position weight; low intensity decreases it.

DWCS is a designed assumption, not an empirically validated constant. This is acknowledged explicitly in the framework documentation. Future versions should validate the DWCS multipliers against five-year portfolio outcome data — a recommendation captured in Chapter 8.

5.7 Scenario Valuation Model

The valuation layer is structurally separate from the scoring layer. The CRS asks "how commercially ready is this startup?"; the valuation model asks "what is it worth, under what scenarios, and by which method?" The two are linked through the Stage-to-Method bridge described in Table 5.5.

Bull, Base, and Bear scenarios are constructed not as arbitrary sensitivities but as coherent narrative cases:

Bull case (Optimistic). Technology scales as planned; market adoption matches projections; no major competitor enters; revenue meets management projections. This is the upside-bound valuation.

Base case (Moderate). Technology scales with 20–30% delay; market adoption at 60–70% of projection; one major competitor enters; revenue at 65% of management projections. This is the most-likely valuation.

Bear case (Pessimistic). Technology faces scale-up challenges; market adoption is slow; regulatory delay or pricing pressure; revenue at 30–40% of management projections. This is the downside-protection valuation.

A worked example illustrates the model's discipline. Suppose a founder claims a ₹50 Cr Year-3 revenue at a ₹150 Cr post-money. The framework applies risk discounts: technology risk –20%, market adoption risk –15%, execution risk –10%, competition risk –10% — a cumulative –55%. Risk-adjusted Year-3 revenue therefore becomes ₹22.5 Cr. At a sector-appropriate 3× revenue multiple, the framework-implied valuation is ₹67.5 Cr — less than half the founder's ask. This is the kind of disciplined counter-narrative that allows EIL to negotiate from a defensible position.

Table 5.5 Stage-to-method bridge for valuation

Stage / CRS Band	TRL Guard	Primary Method	Rationale
≥ 85 (Investment-grade)	TRL ≥ 7	Risk-adjusted DCF + Revenue Multiple	Cashflows proven; multiples sanity-check.
70–84 (Commercially ready)	TRL ≥ 5	VC Method + Strategic Value Uplift	Pre-revenue but credible growth horizon.
58–69 (Promising)	Any	Scorecard Method	Limited financial data; benchmark-anchored.
< 58 (Early/Concept)	Any	Berkus Method	Milestone-based pre-revenue floor.

Sector-specific terminal multiples (Table 5.6) anchor DCF and VC Method outputs to observed exit reality rather than a single tech-grade default. Three of the fourteen multiples — CCUS, nano, and wind — were adjusted in the calibration phase to reduce variance against actual transactions; the diagnostic is in Section 5.11.

Table 5.6 Sector-specific exit multiples (calibrated)

Sector	Multiple	Sector	Multiple
Green Hydrogen	8×	IoT / Smart Grid	10×
Solar / PV	5×	Wind (calibrated)	3×
Energy Storage	7×	Nuclear / SMR	12×
CCUS (calibrated)	12×	Nanomaterials (calibrated)	9×
Biofuels	5×	Waste-to-Energy	5×
AI Energy Efficiency	8×	Water Tech	6×
Industrial Biotech	7×	Other Deep-Tech	6×

5.8 EIL Action Recommendation Engine

The recommendation engine maps a startup's combination of CRS score, hard-floor gates, and DWCS-adjusted context to one of four actionable pathways: Equity, Licensing, Engineering Partnership, or Reject (with Watch as a holding pattern). Each pathway carries distinct downstream implications for EIL: Equity implies a board seat, pro-rata rights, and an information covenant; Licensing implies an upfront royalty plus running-royalty structure; Engineering Partnership implies a paid technology-integration agreement within an EPC scope; Reject closes the file. The decision logic is encoded in the platform's report module and is surfaced as the final card on the Report page.

5.9 Digital Platform Development

The framework is operationalised through a browser-based platform deployed at *engsui-evaluator.netlify.app*. The platform was deliberately implemented as a pure HTML/CSS/JavaScript single-page application — no backend, no database, no API dependencies — for three reasons. First, EIL's IT-procurement norms make server-hosted internal tools slow to onboard.

Second, a client-side application carries zero data outside the user's browser, eliminating data-residency concerns for sensitive startup financials. Third, the platform can run inside an air-gapped EIL workstation if needed.

The platform comprises eleven screens organised under a persistent tab navigation: Profile, Modules M1 through M7, Valuation Configuration, Data Room, and Report. The Profile screen captures basic startup metadata and manages the saved portfolio. The seven module screens host the sub-indicator questionnaires with inline evaluator notes. Valuation Configuration exposes tunable parameters (WACC, terminal multiple, TRL discount, IRR, exit horizon, sector multiple overrides). The Data Room captures due-diligence inputs (CIN, DPIIT recognition status, cap table, prior round terms, EHS notes). The Report screen renders the full evaluation output — CRS breakdown, DWCS overlay, Bull/Base/Bear valuation cards, sensitivity tables, advisory flags, and engagement recommendation.

Several non-obvious design decisions improve the platform's operational usefulness. *Evidence confidence flags* at module level allow evaluators to record High/Medium/Low confidence for each module score; the report surfaces "high score + low confidence" combinations as risk flags for further diligence. *Demo-data seeding* on first visit pre-loads eight calibration startups so that the tool is usable immediately without manual data entry. *Trash retention* with a 30-second undo window and a five-day retention period protects against accidental deletion of in-progress evaluations. *JSON export/import* enables portfolio sharing without a backend. The deployment uses Netlify with secure-header rules and a 301 redirect from legacy paths to the canonical scoring URL.

The platform was deployed on 1 June 2026 and is accessible at three URLs: engsui-evaluator.netlify.app (main tool), engsui-evaluator.netlify.app/scoring-mechanism/ (full 120-point hierarchy reference), and engsui-evaluator.netlify.app/guide/ (functioning guide).

5.10 Primary Research — Founder Interviews

Six interviews were conducted between Weeks 5 and 7 of the internship. The instrument is reproduced as Appendix B. Each interview was a structured forty- to seventy-five-minute call, coded into the platform within twenty-four hours, and cross-referenced against publicly disclosed data prior to scoring. Table 5.7 summarises coverage.

Table 5.7 Founder interview coverage (six startups)

Startup	Interviewee	Sector	Stage	Module Focus
Fermentech Labs	Dr. Sidharth Arora (Founder/CEO)	Industrial Biotech	Pre-seed	Tech, IP, Market, EIL Fit
Powerzest Energy Solutions	Mr. Avinash Gupta (Co-founder)	Micro-renewables / Wind	Seed	Tech, Adoption, Business Model, EIL Fit
Aquafront Infrastructure	Mr. Achin Agrawal (Co-founder)	Water / Floating Infra	Seed	Market, Adoption, EPC Fit
E-Spin Nanotech	Dr. Sandip Patil (Founder/CEO)	Nanomaterials	Series A (bootstrapped)	Tech, IP, Market, Business Model
VayuJal Technologies	Mr. Ramesh Kumar (CEO)	Atmospheric Water Generation	Seed	Tech, Market, EIL Fit
Alt Carbon	Mr. Sparsh Agarwal (Co-founder)	CCUS / CDR (ERW)	Seed (Frontier-backed)	Tech, MRV credibility, EIL Fit

5.10.1 Cross-Interview Patterns

Four patterns emerged consistently across interviews. *First*, founders consistently underestimate the time-and-capital required from first commercial pilot to bankable repeatable installation — typically by a factor of two. The framework's Adoption Feasibility module compensates for this by anchoring the scoring on independent benchmarks rather than founder projections. *Second*, every founder interviewed valued EIL's EPC-network access more highly than the funding amount itself, suggesting that Engineering-Partnership pathways may be underused relative to Equity in the current EngSUI deployment mix. *Third*, Chinese competition was named as the most material adoption barrier by three of six founders, particularly in

micro-renewables and nanomaterials — a structural rather than execution issue that the framework now captures through the Competitive Intensity sub-indicator. *Fourth*, regulatory uncertainty (CPCB approvals, BIS certifications, Carbon-credit registry recognition) was named more frequently than technical risk as the proximate scale-up barrier — reinforcing the framework's decision to give Adoption Feasibility a dedicated fifteen-point weight rather than rolling it into Market Validation.

5.10.2 Selected Founder-Level Findings

Fermentech Labs. Solid-state fermentation derived cellulase plus xylanase for 2G ethanol — an EIL-EPCM- adjacent technology with credible non-dilutive funding (DBT BIG, BIRAC, EngSUI grants). Founder articulated a clear two-year path to commercial pilot with a 2G ethanol producer; main constraint is paid-pilot conversion. Framework outcome: CRS ~55%, engagement = Seed Grant or License pathway.

Powerzest Energy. Adaptable Disc Turbine for low-wind generation; patented in India. Founder open about pre- commercial stage and the need for an anchor customer. Pre-revenue with thin team. Framework outcome: CRS ~40%, engagement = Watch with a possible Innovation Challenge pathway.

Aquafront Infrastructure. Floating CNG station on the Ganga commissioned for GAIL; seven-plus-year operational reference at Hirakud and Beohari Dam. The strongest EPCM-adjacent fit in the interview set. Framework outcome: CRS ~70%, engagement = Engineering Partner.

E-Spin Nanotech. Fifteen-year bootstrapped operator with 500+ machine installations globally and DSIR- recognised R&D. Strong execution, modest IP moat, indirect EIL fit. Framework outcome: CRS ~77%, engagement = Commercially-ready / License consideration.

VayuJal Technologies. Atmospheric water generation, IIT Madras spin-out, EIL+HDFC Capital and RTBI listed as supporters. Framework outcome: CRS ~62%, engagement = Equity (validates EIL's prior exposure as a sound first call).

Alt Carbon. First Indian Frontier-backed CDR startup using Enhanced Rock Weathering on Darjeeling tea plantations. Strong EIL alignment in the energy-transition thesis but early business maturity and thin IP moat. Framework outcome: CRS ~61%, engagement = Engineering Partner or License.

5.11 Calibration and Validation

The framework was calibrated against eight live transactions drawn from across the EngSUI-adjacent universe: the six interviewed startups plus two additional cases (Log9 Materials as a positive late-stage case and Miko Technologies as a negative non-EIL-fit case). The calibration approach measured framework-implied valuation against publicly disclosed or dilution-implied market valuations and recorded the variance with a transparent diagnostic tag.

Table 5.8 Calibration summary (eight startups, June 2026)

Startup	Sector	Actual (₹ Cr)	Tool (₹ Cr)	Method	Variance	Validation
Log9 Materials	Storage	1,899.7	1,747	Peer Multiple	+8.7%	Tight ✓
Aquafront Infra	Water	15.0	15.75	Risk-adj DCF	-4.8%	Tight ✓
Fermentech Labs	Biotech	15.0	16.0	Risk-adj DCF	-6.3%	Tight ✓
VayuJal	Water (AWG)	2.0	2.4	Risk-adj DCF	-16.7%	Tight ✓
E-Spin Nanotech	Nanomaterials	300.0	248.0	DCF / Peer	+21.0%	Acceptable △
Powerzest Energy	Wind	0.75	1.2	Risk-adj DCF	-37.5%	Acceptable △
Miko Technologies	IoT (consumer)	4,600	3,071	Mixed	+49.8%	Wide △
Alt Carbon	CCUS / CDR	398.5	220.3	VC Method	+80.9%	Broken ✗

Five of eight cases land inside a twenty-percent variance band — a useful baseline for an internally-owned framework that does not aim to replicate market premium-for-narrative behaviour. The remaining three cases each carry a different diagnostic narrative.

5.11.1 Variance Diagnosis and Parameter Adjustments

Table 5.9 Variance diagnosis and parameter adjustments

Case	Diagnosis	Action Taken
Alt Carbon (+80.9%)	Frontier-backed CDR commands a hot-category premium that the original 9× CCUS multiple did not capture. VC-Method exit value undershot accordingly.	CCUS sector multiple raised from 9× to 12×. Resulting framework valuation ≈ ₹383 Cr vs actual ₹398.5 Cr — variance now ~4%.
Miko Technologies (+49.8%)	Series-D consumer-AI robotics with category-leader narrative premium. Framework deliberately not tuned for non-EIL-fit cases.	No parameter change. Miko is a negative case included to validate the EIL Strategic Fit hard floor — confirmed working (CRS 79% but engagement = Reject).
E-Spin Nanotech (+21%)	Nanomaterials terminal multiple under-weighted for execution-strong bootstrapped operator with global installed base.	Nano sector multiple raised from 8× to 9×. Resulting framework valuation ≈ ₹278 Cr vs actual ₹300 Cr — variance now ~7%.
Powerzest Energy (-37.5%)	Pre-revenue micro-renewables overshoot by default 35%/6× DCF tier; embryonic-stage downside not adequately discounted.	(i) Wind sector multiple lowered from 4× to 3×; (ii) new "embryonic" DCF tier introduced (TRL ≤ 4 with arr0 = 0 → 40% WACC, 4× terminal). Resulting framework valuation ≈ ₹0.8 Cr vs actual ₹0.75 Cr.

The mean valuation ratio across the eight cases — framework valuation divided by market valuation — settles at approximately 0.59× after the calibration adjustments. This ratio is best interpreted not as inaccuracy but as *deliberate conservatism*: the framework systematically excludes narrative premium and category momentum from its baseline output. For a PSU

strategic investor whose mandate is deployment value rather than financial upside, a conservative number is the correct output. The framework allows the analyst to identify when the market has paid a premium and to explicitly justify or discount that premium — rather than embedding it implicitly in the model.

5.11.2 Failure-Case Validation: Log9 Materials

Log9 Materials was incorporated as a retrospective failure-case validation. Public reporting on Log9's FY24 financials shows ARR of ₹110.3 Cr against an EBITDA margin of -67% and a debt-to-equity ratio of 1.68 — i.e. a fast-growing topline funded by deteriorating unit economics. Applied retrospectively to the Log9 inputs as of its Oct 2023 Series B (the round at which the ~₹1,900 Cr post-money was set), the framework returns: CRS ~75% (Commercially ready); Adoption Feasibility flagged as low due to Chinese-competitor pressure on cell costs; Business Model flagged as low due to negative EBITDA at ARR scale. The recommended engagement is License or Engineering Partner, *not* Direct Equity. In other words, the framework would have surfaced the structural risks that materialised in Log9's subsequent operational difficulties. This is the strongest single piece of validation in the calibration exercise.

5.12 Portfolio-Level Synthesis and EIL Role Matrix

A framework that only produces individual evaluations is half-finished. The investment committee operates over portfolios. The final synthesis converts the eight calibration cases into an EIL Role Matrix that tells the committee, for each startup, exactly what role EIL is best placed to play.

Table 5.10 Portfolio EIL role matrix

Startup	CRS	Recommended Engagement	EIL's Best Role
Log9 Materials	75%	License / Engineering Partner	Battery integration partner for stationary-storage projects
E-Spin Nanotech	77%	License	Filtration technology licensor for refinery HVAC and emission-control retrofit
Aquafront Infra	70%	Engineering Partner	Floating infrastructure EPC partner; government-tender facilitator
VayuJal	62%	Equity (validates prior)	Anchor PSU customer for AWG units at remote project sites
Alt Carbon	61%	Engineering Partner	MRV / regulatory integrator; carbon-credit offtake intermediary
Fermentech Labs	55%	Seed Grant / License	Industrial customer connector for 2G ethanol producers
Powerzest Energy	40%	Watch	Pre-investment introduction to PSU client base; not equity
Miko Technologies	79%	Reject (Hard floor)	Out of EIL mandate — no role

Two patterns are visible. First, engagement pathways concentrate in License and Engineering Partnership rather than Equity. This is consistent with the founder-interview finding that EIL's network is the most-valued asset. Second, the framework correctly identifies the single non-EIL case (Miko) regardless of its high CRS — confirming the strategic-fit hard floor as the most consequential single design decision.

CHAPTER 6 — EXPERIENCE GAINED AND PROBLEMS FACED

6.1 Key Learnings

Eight weeks of structured exposure to a PSU investment process surfaced learnings of three kinds: substantive learnings about deep-tech valuation, organisational learnings about PSU decision behaviour, and methodological learnings about how to build frameworks under uncertainty.

On substance: deep-tech valuation differs from consumer-startup valuation in three ways that the academic literature does not fully capture. Cash-flow horizons are longer; capital intensity is higher; and the meaningful exit multiples are sector-specific rather than tech-grade. A framework that treats deep-tech as a sub-genre of "early-stage venture" will systematically misprice the portfolio. The framework here is built around this distinction from the start.

On organisations: PSU investment decision-making operates under constraints — audit defensibility, mandate boundaries, governance process — that meaningfully change what a "useful" framework looks like. A framework that maximises IRR but cannot be defended in front of the CAG is unusable. A framework that produces a modest internal rate of return but is fully traceable to evidence is preferred. The deployed platform reflects this with its evidence-confidence flags, sub-indicator-level notes, and auditable JSON exports.

On methodology: weight calibration is a question of acknowledged judgement, not empirical truth. Both the CRS module weights and the DWCS multipliers are defended in the framework documentation as deliberate assumptions, not as constants extracted from data. This honesty is unusual in framework documents but seems to be the right stance for an internally-owned tool.

6.2 Challenges Faced

Four challenges materially constrained the internship.

Confidentiality of TC reports. TC reports — the most directly useful comparative input — could not be cited verbatim, and only limited de-identified content could be referenced. This forced the framework to be designed from publicly inferable patterns rather than from confidential ground truth. The resulting framework is more defensible (no derivative dependency on TC content) but less data-rich than it could otherwise have been.

Weight calibration without empirical dataset. Module weights ideally would be empirically derived from outcome data across past EngSUI investments. The portfolio is, however, too young and too small to support such an exercise: the oldest cohort funded in 2017–18 has not fully matured to outcome, and sample sizes within any sector are in single digits. The fallback was to anchor weights on startup-failure-mode literature (NESTA, CB Insights, Sequoia) and to be explicit about the assumption nature of the weights.

Scheduling founder interviews. Co-ordinating six interviews across two months within a calendar populated with founder fundraising commitments was the single largest operational constraint. Each interview required two to three rescheduling attempts. The success of the calibration layer depended materially on the patience of the Manager (R&D), who facilitated the introductions.

IP module rigour. The IP module relies on sub-indicators (claim breadth, claim quality, FTO) that properly require patent-counsel input. The framework acknowledges this limitation: IP scores above ten are flagged in the report as warranting external counsel verification before use in a transaction document.

6.3 Areas for Improvement

The framework as delivered would benefit from four extensions in future iterations:

1. *Empirically validated DWCS multipliers.* Once EngSUI's portfolio has matured to a sufficient number of outcomes, the DWCS multipliers should be back-tested against actual outcomes and refit. The current values are defended as reasoned assumptions, not as constants.

2. *Patent-counsel review integration.* A formal interface for external patent-counsel review of the IP module, with structured input fields rather than free-text notes.
3. *Confidence interval rather than point estimate.* The valuation engine produces three scenario points (Bull, Base, Bear). A future version should produce a full probability distribution across these scenarios using a Monte Carlo overlay.
4. *Multi-user collaboration.* The current platform is single-user. A future version should support multi-evaluator scoring with disagreement surfacing, particularly useful for investment-committee preparation.

6.4 How the Experience Helps Future Career

The internship's most transferable contribution to career trajectory is exposure to the way real investment decisions get made under institutional constraint. The textbook DCF taught in Corporate Finance is a starting point, not a destination. What the investment committee actually needs is a framework that produces a defensible number, an auditable rationale, and an explicit recognition of where judgement has been exercised. Building such a framework — and operationalising it as a working tool — is a skill applicable directly to consulting strategy roles, to product management in fintech and climate-tech, and to corporate development positions in any technology-integrating enterprise.

6.5 Internship Work Plan and Actual Progress

Table 6.1 Internship work plan and actual progress

Week	Planned Activity	Status	Output
1	EIL orientation; EngSUI study; TC report review	Completed	Project scope defined and signed off
2	Framework architecture design; literature review	Completed	Seven-module CRS designed; weights justified
3	Platform development v1; scoring hierarchy build	Completed	Interactive platform v1 deployed locally
4	Calibration testing v1; framework refinement	Completed	Three-startup calibration; weights tightened
5	Founder interviews v1 (Fermentech, Powerzest)	Completed	Interview notes coded; CRS scores generated
6	Founder interviews v2 (Aquafront, E-Spin, VayuJal, Alt Carbon)	Completed	Interview notes coded; portfolio matrix built
7	Calibration v2 (8 startups); platform refinement	Completed	Calibration CSV produced; sector multiples re-tuned
8	Platform deployment on Netlify; final report writing	Completed	Platform live at engsui-evaluator.netlify.app ; final report submitted

CHAPTER 7 — LEARNINGS AND UTILISATION OF MBA COURSES DURING SIP

The IIM Nagpur first-year MBA curriculum was instrumental in several specific framework design decisions. The mapping below is deliberately precise: each subject is mapped not to "general relevance" but to a specific framework element where its concepts were applied.

Table 7.1 MBA subject-to-framework mapping

Subject	Concepts Used	Framework Application
Corporate Finance / Financial Management	DCF, NPV, WACC, terminal value, sensitivity analysis	Risk-adjusted DCF tiering (mature / projecting / embryonic); sensitivity matrix in the valuation report; Bull/Base/Bear scenario construction
Business / Strategic Management	Porter's Five Forces; VRIO; competitive advantage	M4 Market Validation (competitive intensity); M5 Adoption Feasibility (incumbent disruption); EIL Strategic Fit module design
Marketing Management / Market Research	TAM / SAM / SOM; primary research design; customer discovery	Market sizing sub-indicators; thirty-two-question structured interview instrument; LOI / paying-customer scoring
Entrepreneurship / Venture Capital	Berkus, Scorecard, Risk-Factor Summation; pre-seed mechanics	Stage-to-Method bridge for valuation; Berkus floor for TRL 1–3; Scorecard adjustment for Promising-band cases
Organisational Behaviour	Stakeholder mapping; PSU decision dynamics	EIL Role Matrix design; recommendation-engine framing as "EIL-best-role" rather than generic "investment"
Operations Management	Systems thinking; process flow design	Eleven-screen platform architecture; module-by-module evaluation pipeline; report generation pipeline
Statistics / Quantitative Methods	Weighted scoring; sensitivity analysis; variance diagnostics	DWCS formula; calibration variance tagging; valuation-method recommendation logic
Business Communication	Structured writing; consulting deliverable design	Final report structure; platform documentation; investment-committee-ready report generation

Three subjects deserve specific mention. **Corporate Finance** provided the technical foundation for the DCF and VC Method implementations; the embryonic tier introduced after calibration is itself a direct application of the lecture on WACC sensitivity to stage. **Strategic Management** provided the conceptual scaffolding for distinguishing strategic fit from financial fit — the single most consequential design decision in the framework. **Marketing Management**'s treatment of customer-discovery rigour informed the design of the thirty-two-question interview instrument and the LOI scoring logic in Market Validation.

Equally important were the unspoken pedagogical contributions: the standard of writing demanded by case-method classes, the norms of evidence-anchored argument, and the consulting-style discipline of structuring recommendations against an explicit framework. These are not specific concepts but they are the medium through which the framework is delivered.

CHAPTER 8 – CONCLUSION, SUMMARY AND RECOMMENDATIONS

8.1 Summary of Work Completed

Over eight weeks the internship has delivered a coherent bundle spanning framework, tool, and validation: (i) the Commercialisation Readiness Scorecard — a seven-module, 120-point evaluation framework with explicit hard-floor gates; (ii) the Dynamic Weight Calibration System — a stage- and context-sensitive overlay on top of the CRS; (iii) the Scenario Valuation Model — a stage-to- method bridge that selects among Berkus, Scorecard, VC Method, Revenue Multiple, Peer Multiple, and risk-adjusted DCF, and produces Bull/Base/Bear scenarios; (iv) the deployed digital platform at *engsui-evaluator.netlify.app*; (v) the eight-startup calibration corpus with transparent variance diagnostics; (vi) the six-founder structured interview programme; and (vii) the EIL Role Matrix that converts individual evaluations into portfolio-level recommendations.

8.2 Key Findings

Four findings emerge from the body of work.

First, the framework's mean valuation ratio of approximately 0.59× market is not inaccuracy but a deliberate feature. For a PSU investor whose mandate is strategic deployment rather than upside maximisation, a systematically conservative output is the correct primary number. The framework supports explicit identification and (where defended) inclusion of narrative premium — but it never embeds it implicitly.

Second, the EIL Strategic Fit module's hard- floor gate functions exactly as designed. Miko Technologies — a consumer-AI robotics company with strong execution, a high CRS, and a recent Series-D round — is correctly rejected by the framework on EIL-fit grounds despite scoring 79% on the general scorecard. This is the single most important calibration result.

Third, the founder interviews reveal that EIL's EPC network and PSU-client relationships are valued more highly by founders than the funding itself. This argues for a portfolio composition that shifts further toward Engineering Partnership and Licensing pathways relative to Direct Equity.

Fourth, calibration variance is sector- structured, not random. CCUS, nanomaterials, and pre-revenue wind each required specific parameter adjustments. This argues for a future practice of running a calibration pass at the start of each new sector entry, rather than treating sector multiples as immutable.

8.3 Recommendations to EIL

Five concrete recommendations follow from the internship.

- 1. Formalise the CRS platform as the standard pre-screen for EngSUI applications.** The platform replaces an ad-hoc pre-screen with a structured one and produces an auditable artefact in JSON form. Cost of adoption is near-zero (no infrastructure, no licence fees).
- 2. Make DPIIT recognition a prerequisite for the Direct Equity track.** DPIIT recognition removes Angel-Tax exposure on premium-priced equity rounds and signals minimum institutional discipline. The cost is trivial; the audit benefit is large.
- 3. Embed pro-rata participation and information rights in all future MOAs.** Even where EIL's initial commitment is sub-board-seat, pro-rata rights preserve optionality on follow-on rounds, and standard information covenants enable portfolio monitoring without renegotiation. Both are standard institutional-investor protections that EIL should make non-negotiable.
- 4. Institute mandatory IP-counsel review for commitments above ₹5 Cr.** The IP module's sub-indicators on claim breadth and FTO require patent-counsel input that the framework cannot substitute for. A formal counsel-review gate above the ₹5 Cr threshold mitigates downstream IP risk.
- 5. Build a lightweight API bridge to MCA21 and the Startup India portal.** Automated compliance verification — incorporation, DPIIT recognition currency, AGM filing — eliminates a category of manual diligence and reduces turnaround time. The investment is modest and amortises across the entire EngSUI pipeline.

8.4 Future Plan and Continuity

Three follow-on workstreams would strengthen the deliverable further within EIL's next eighteen months:

(i) *Portfolio outcome backtest* — once the 2017–2021 cohort matures to operational outcome, run the framework against historical inputs and measure score-to-outcome correlation. This is the empirical foundation for refining the DWCS multipliers. (ii) *Multi-evaluator collaboration on the platform* — extend the platform to support investment-committee preparation workflows where multiple evaluators score independently and disagreement is surfaced for discussion. (iii) *Sector-pack extensions* — pre-populate sector-specific sub-indicator thresholds (e.g. for green hydrogen, CCUS, biofuels) so that new applicants can be screened against a sector-tuned rubric rather than the generic one.

8.5 Concluding Statement

EIL's EngSUI initiative occupies an unusually consequential institutional space — at the intersection of national energy policy, public-sector procurement discipline, and the operating realities of early-stage deep-tech. A framework appropriate to this space cannot be borrowed wholesale from generalist venture capital practice, nor can it be derived solely from grant-funding rubrics like BIRAC's. It must be built ground-up around EIL's strategic mandate, calibrated against actually-observed transactions, and operationalised as a tool the analyst can actually use on a Tuesday afternoon.

The Commercialisation Readiness Scorecard, the Dynamic Weight Calibration System, the Scenario Valuation Model, the deployed platform, the eight-startup calibration corpus, the six-founder interview programme, and the EIL Role Matrix together attempt to meet this standard. The internship has produced a structured, academically grounded, technically deployed, and empirically calibrated capability that addresses a genuine gap in EIL's investment-decision process. The framework bridges the gap between the TC's numerical valuation and EIL's strategic investment rationale — providing a repeatable, auditable, and internally owned evaluation capability that is now operational and available for use by the R&D Division. The author

thanks Engineers India Limited for the opportunity to contribute to this work and looks forward to its continued use and refinement in the EngSUI initiative's future funding rounds.

APPENDIX A — DETAILED SCORING INDICATOR MAP

The appendix below enumerates the 40+ sub-indicators that the CRS framework uses across its seven modules. Each sub-indicator's point allocation and the scoring rubric (1–5 scale) is documented in the platform's Methodology page at engsui-evaluator.netlify.app/scoring-mechanism/.

M1 — Technology Readiness (25 points)

- **TRL Level (5 pts)** — European-Commission nine-level scale; benchmarked to industrial maturity gates.
- **Pilot Status (5 pts)** — paid pilot, unpaid pilot, controlled lab, none.
- **Independent Validation (2 pts)** — third-party endurance / performance verification by an accredited laboratory.
- **Data Quality (3 pts)** — depth and credibility of performance data disclosed.
- **Uniqueness (3 pts)** — defensibility of technological position vs nearest competitor.
- **Scalability (3 pts)** — engineering replicability from prototype to commercial scale.
- **Manufacturing Maturity (2 pts)** — readiness of manufacturing process and supply chain.
- **Reliability (2 pts)** — MTBF, failure-mode analysis, field experience.

M2 — IP and Patents (15 points)

- **Filing Status (3 pts)** — grants vs filings vs provisional.
- **Breadth (2 pts)** — geographic and claim coverage.
- **Claim Quality (3 pts)** — independent-claim defensibility (requires patent-counsel input).
- **FTO Status (3 pts)** — freedom-to-operate clearance.
- **Publications (1 pt)** — peer-reviewed scientific support.
- **Origin (2 pts)** — ownership clarity (founders vs institution vs jointly assigned).
- **Trade-Secret Layer (1 pt)** — protection of process know-how outside the patent envelope.

M3 — Team and Leadership (20 points)

- **Founder Domain Expertise (4 pts).**
- **Business / Commercial Depth (3 pts).**
- **Industry Network and Relationship Capital (2 pts).**
- **Team Completeness (3 pts)** — coverage of CTO/CFO/COO functions.
- **Institutional Anchoring (3 pts)** — IIT/IISc/NIT or PhD-program origin.
- **Diversity and Adaptability (2 pts).**
- **Advisory Board (1 pt).**
- **Prior Traction / Exit (1 pt).**
- **Full-Time Intensity (1 pt).**

M4 — Market Validation (20 points)

- **TAM (3 pts), SAM (3 pts), CAGR (2 pts).**
- **Customer Discovery Depth (2 pts).**
- **LOI / Pilot Evidence (3 pts).**
- **Paying Customers (2 pts)** — binary discount removed if zero.
- **Unit-of-Consumption Fit (2 pts).**
- **Competitive Intensity (Inverse, 2 pts).**
- **Policy Tailwind (1 pt).**

M5 — Adoption Feasibility (15 points)

- **Regulatory Path (3 pts).**
- **CapEx Intensity (Inverse, 5 pts).**
- **Integration Complexity (Inverse, 2 pts).**
- **Incumbent Disruption (2 pts).**
- **Supply Chain Readiness (2 pts).**
- **Price Competitiveness (1 pt).**

M6 — Business Model (15 points)

- **Revenue-Model Clarity (3 pts).**

- **Unit Economics (4 pts).**
- **Customer Acquisition Strategy (2 pts).**
- **Operating Discipline (2 pts).**
- **ARR Trajectory (2 pts).**
- **Runway (2 pts).**

M7 — EIL Strategic Fit (10 points)

- **Sector Alignment (3 pts).**
- **EPC Integration Potential (3 pts).**
- **National-Mission Fit (2 pts).**
- **Licensing Willingness (1 pt).**
- **Founder Receptivity (1 pt).**

APPENDIX B — FOUNDER INTERVIEW QUESTIONNAIRE

The instrument below was used in all six founder interviews conducted in Weeks 5–7. The instrument is structured across five modules of approximately ten minutes each, with one wrap-up module of five minutes — total approximate duration forty-five minutes.

Module 1 — Technology (10 minutes)

1. Walk me through where your technology stands today — lab, pilot, or commercial scale.
2. What is the single biggest technical risk that could prevent scale-up?
3. Patents filed or granted? What exactly is protected?
4. What time and capital does it take to go from your current stage to first commercial deployment?
5. Has the technology been independently validated? By whom?

Module 2 — Market and Customers (10 minutes)

1. Who is the ideal customer, and have you spoken to them?
2. Any paying customers, LOIs, or pilots underway?
3. The biggest reason a potential customer has said "no" or delayed?
4. Top three competitors — including Chinese or international.
5. Pricing model and the basis for it.

Module 3 — Financials and Scale (10 minutes)

1. Current production cost per unit at pilot scale.
2. Estimated cost per unit at commercial scale.
3. Capital required to reach commercial scale — and what it is used for.
4. Revenue model (product sale, licence, SaaS, service)?
5. Projected payback period for the customer investing in your technology.

Module 4 — EIL Partnership (10 minutes)

1. Beyond funding, what does EIL's network or expertise offer you?

2. Specific industries or companies where EIL's relationships could open doors?
3. Have you explored licensing — where EIL or its clients license your technology?
4. Ideal exit scenario — acquisition, IPO, licence revenue?
5. The one thing EIL could do that would most accelerate your commercialisation.

Module 5 — Risks and Barriers (5 minutes)

1. Biggest non-technical barrier to adoption in your sector.
2. Regulatory approvals pending that could delay commercialisation.
3. What happens if a large player (Reliance, Tata, ONGC) decides to build this themselves?

APPENDIX C — CALIBRATION DETAIL AND METHODOLOGY

The calibration approach measures framework-implied valuation against publicly disclosed or dilution-implied market valuations and records the variance with a transparent tag. The methodology is described in Section 5.11 of the main report; this appendix provides the detailed per-startup table.

Appendix C.1 Per-startup calibration detail

Startup	Actual (₹ Cr)	Actual-Value Basis	Tool (₹ Cr)	Tool Method	Tool Verdict
Alt Carbon	398.5	\$12M Seed May 2025 @ 25% dilution- implied	220.3 (pre) / ≈383 (post- calib)	VC Method	Promising (~61% CRS)
Miko Technologies	4,600	Series D Aug 2025 · \$550M post-money	3,071 (pre) / not re- tuned	Mixed	Commercially- ready CRS but engagement = Reject (non-EIL fit)
E-Spin Nanotech	300	Bootstrapped 15-yr operator (founder ask ~₹20 Cr)	248 (pre) / ≈278 (post- calib)	Risk-adj DCF / Peer Mult	Commercially- ready (~77%)
Log9 Materials	1,899.7	Series B Oct 2023 · \$228M post-money	1,747	Peer Multiple	Commercially- ready (~75%) · License / Equity
Aquafront Infra	15.0	\$0.5M Seed FY23-24 @ 20–30% dilution- implied	15.75	Risk-adj DCF (water 6× terminal)	Commercially- ready (~70%) · EPC Partner
Fermentech Labs	15.0	Bootstrapped · DBT BIG + BIRAC + EngSUI grants	16.0	Risk-adj DCF (Biotech 7×)	Promising (~55%) · Seed Grant / License
VayuJal	2.0	RTBI 5% via incubator grant · no priced equity round	2.4	Risk-adj DCF (Water 6×)	Viable niche (~62%) · Equity (validates EIL prior)
Powerzest Energy	0.75	TIDES incubator	1.2 (pre) / ≈0.8	Risk-adj DCF	Early/ exploratory (~40%) · Watch

		only · no equity round	(post- calib)	embryonic tier	
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Variance is recorded as $(\text{Tool} - \text{Actual}) / \text{Actual} \times 100$. Negative variance indicates framework overshoot relative to actual; positive variance indicates undershoot. The five tight cases all land inside a twenty-percent variance band. The three wider cases each carry a documented diagnostic narrative; two (Alt Carbon, E-Spin) were addressed via sector-multiple recalibration and one (Powerzest) via a new embryonic DCF tier. The Miko case was deliberately left un-tuned to validate the EIL Strategic Fit hard floor.

APPENDIX D – INTERNSHIP OFFER LETTER (EIL)

The original allocation email from Engineers India Limited is reproduced below. The original electronic copy is retained on file at IIM Nagpur's CDS office; the printed hard-bound copy of this report carries the original e-mail print-out as the official offer document.

Subject: Reporting of Vocational Trainee Mr. Priyanshu Dutta in EIL from 05-April-2026 to 06-June-2026

From: Sudhansu Bala Satapathy <sb.satapathy@eil.co.in>

To: Sanjeev Singhal <sanjeev.singhal@eil.co.in>

Cc: Anubha T. Agarwal, p25priyanshu@iimnagpur.ac.in, Anil Kumar, trainingdepartmenteil@gmail.com

Date: Thursday, 5 March 2026 at 3:36 PM

Dear Sir,

Below mentioned VT(s) has/have been allocated to you for Internship as per details below. CV of the student(s) are enclosed for reference.

Table D.1 Allocation details from EIL Training & Development

Student Name	Email	Contact	Division	Location	Reporting
Mr. Priyanshu Dutta	p25priyanshu@iimnagpur.ac.in	7043373525	R&D	Gurugram Complex	Mr. Sanjeev Singhal

Mr. Priyanshu Dutta: Please share attached Bio-data form by 5 April 2026 to Ms. Sudhansu Bala Satapathy at sb.satapathy@eil.co.in. For joining formalities, the student is to report to the undersigned at Training & Development Division, EIL, Gurugram, at 8:30 a.m. on 5 April 2026 carrying a printout of this e-mail, a copy of the Institute ID Card, two passport- size photographs, Aadhaar ID, and the filled Bio-data form.

The student will submit the Evaluation & Attendance proforma (hard copy, to be provided at the time of joining) after it is duly filled and approved, on the last day of training.

Regards,

S. B. Satapathy

Training and Development

Tower II, G-Floor · EIL Gurugram Office Complex

Phone: 0124-289-2242 · Mobile: 9953947089

[The original e-mail print-out is bound at the front of the hard-bound copy of this report along with the EIL Summer Internship Completion Certificate issued by the R&D Division at the close of the internship.]

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— End of Report —